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mcSCRB-seq2 protocol

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Protocol status: Working

We used this protocol and it's working.

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Abstract

mcSCRB-seq2 is the improved version of our published protocol molecular crowding SCRБ-seq ([mcSCRБ-seq](https://www.nature.com/articles/s41467-018-05347-6)) ([https://www.nature.com/articles/s41467-018-05347-6/](https://www.nature.com/articles/s41467-018-05347-6)).

Following changes were introduced:

- re-evaluted the Lysis Buffer formulation, making it more robust against RNAses
- added additional dCTP in the Reverse Transcription leading to higher yield by enhanced Template switching
- updated the oligo dT primers to be compatible with 10x's three prime gene expression layout (see [prime-seq](#) for more information)
- changed to KapaHifi for PreAmplification as it produces less chimeric rerads compared to Tera Polymerase
- increased Primer concentration in PreAmp (SingV6)
- included an option for cheaper library prep with NEBNext

Guidelines

- All reagents and plasticware can be found in the 'Materials' section.
- Use only RNase free supplies and clean all surfaces and tools with RNase Away prior to working
- Make sure all steps before reverse transcription are carried out swiftly and on ice.
- All primer sequences are listed below:

	A	B	C	D	E
	Oligo	Vendor	Purification	Concentration	Sequence
	barcoded oligo-dT (E3V7NEXT)	Sigma	Cartridge	2 µM	ACACTCTTTCCCTACACGA CGCTCTTCCGATCT[BC12] [UMI16][T30]VN
	TSO unblocked (E5V7NEXT)	Sigma	RNase-Free HPLC	100 µM	Biotin- ACACTCTTTCCCTACACGA CGCrGrGrG
	PreAmp (SINGV6)	IDT	Desalted	10 µM	Biotin- ACACTCTTTCCCTACACGA CGC
	3' enrichment primer (P5NEXTPT 5)	IDT	HPLC	5 µM	AATGATACGGCGACCACC GAGATCTACACTCTTTCCC TACACGACGCTCTTCCG*A *T*C*T
	i7 Index Primer (Nextera)	IDT	TruGrade	5 µM	CAAGCAGAAGACGGCATA CGAGAT[i7]GTCTCGTGGG CTCGG
	i5 Index Primer (TruSeq)	IDT	Trugrade	5µM	AATGATACGGCGACCACC GAGATCTACAC[i5]ACACTC TTTCCCTACACGACGCTCT TCCGATCT

Specific barcoded oligodT (E3V7NEXT) sequences:

 E3V7_Set1.txt

 E3V7_Set2.txt



Materials

General:

- ☒ RNase AWAY™ Surface Decontaminant **Carl Roth Catalog #A998.4**
- ☒ Ethanol (100%, Molecular Biology Grade) **Fisher Scientific Catalog #BP2818500**
- ☒ UltraPure™ DNase/RNase-Free Distilled Water **Thermo Fisher Scientific Catalog #10977023**
- ☒ Quant-iT™ PicoGreen® dsDNA Assay Kit **Life Technologies Catalog #P11496** or
- ☒ QuantiFluor(R) dsDNA System **Promega Catalog #E2670**

Homemade Beads:

- ☒ Sera-Mag SpeedBeads Carboxylate-Modified Magnetic Particles **GE Healthcare Catalog #44152105050350**
- ☒ Sodium Chloride **Fisher Scientific Catalog #S271**
- ☒ Tris HCl Buffer 1M Solution, Sterile pH 8.0 **Bio Basic Inc. Catalog #SD8127.SIZE.450ml**
- ☒ EDTA (0.5 M), pH 8.0 **Life Technologies Catalog #AM9260G**
- ☒ IGEPAL-CA630 **Merck MilliporeSigma (Sigma-Aldrich) Catalog #I3021 SIGMA-ALDRICH**
- ☒ Sodium Azide **Merck MilliporeSigma (Sigma-Aldrich) Catalog #S2002-100G**

Lysis:

- ☒ Phusion HF Buffer Pack - 6.0 ml **New England Biolabs Catalog #B0518S**
- ☒ NxGen RNase Inhibitor **Lucigen Catalog #30281-2**

Reverse Transcription:

- ☒ PEG 8000 **Merck MilliporeSigma (Sigma-Aldrich) Catalog #81268**
- ☒ Maxima H Minus Reverse Transcriptase **Thermo Fisher Scientific Catalog ##EP0741**
- ☒ dNTP Set 100 mM Solutions **Thermo Fisher Scientific Catalog #R0181**
- ☒ dCTP Solution (100 mM) **Thermo Fisher Scientific Catalog #R0151**
- ☒ ERCC RNA Spike-In Mix **Thermo Fisher Catalog #4456740** (optional)
- ☒ Exonuclease I (E.coli) - 15,000 units **New England Biolabs Catalog #M0293L**

PreAmp:

- ☒ Kapa HiFi Hotstart ReadyMix (2x) **Kapa Biosystems Catalog #KK2612**

Library:



⊗ Bioanalyzer chips and reagents (DNA High Sensitivity kit) **Agilent Technologies**

⊗ NEBNext Ultra II DNA Library Prep with Sample Purification Beads - 96 rxns **New England Biolabs Catalog #E7103L**

⊗ NEBNext Q5 Hot Start HiFi PCR Master Mix - 250 rxns **New England Biolabs Catalog #M0543L**

⊗ SPRIselect reagent **GE Healthcare Catalog #B23317**

Before start

Wipe bench surfaces with RNase Away and keep working environment clean.

Sample Collection

- 1 Prepare Lysis Buffer for the number of plates you intend to use.

	A	B	C
	Reagent	per well	per plate (96 well)
	Buffer Phusion HF 5x	0.008	0.88
	NxGen RNAi (40U/ul)	0.125	13.75
	H2O	3.867	425.37
	Total	4	440

Lysis Buffer PPI (Primer, Phusion and inhibitor)

- 2 Distribute  4 µL **PPI Lysis Buffer** per well using a repeater pipette (stepper) or similar.

Add  1 µL barcoded oligo-dT primer [2 µM] (E3V7) to each well using a multichannel pipette.

Note

When preparing many plates at once we usually make one **master plate including the primers** and then distribute  5 µL to each well of the Lysis plates.

Lysis plates with barcode primers can be prepared ahead of time and stored at  -20 °C for short term (< 1 week).

- 3 Deposit one cell per well in the Lysis plate using e.g. FACS

Note

Keep plate cooled whenever possible, briefly spin down before use to make sure lysis buffer is at the bottom of the well.



4 Immediately after sorting close plate tightly with an aluminum seal.

5 In a cooled centrifuge, spin down the plate

 1000 x g, 4°C, 00:00:30 , max speed if 1000 x g are not possible and place immediately on dry ice.

Store plates at  -70 °C or lower for up to 6 month before processing.

30s

Preparation

6 Before starting clean all surfaces and pipettes with RNase Away.
Apply RNase Away for 3-5 minutes and wipe away with a clean tissue.

7 When running the protocol for the **first time**, prepare **Cleanup Beads** as described in Appendix 1.

Note

The **Cleanup Beads** can be stored at  4 °C for up to six months and do not have to be prepared fresh every time.

8 When running the protocol for the **first time**, prepare **Pooling Beads** as described in Appendix 2.

Note

The **Pooling Beads** can be stored at  4 °C for up to six months and do not have to be prepared fresh every time. **Pooling Beads** contain a 5 times lower amount of Bead particles and can alternatively be prepared by diluting **Cleanup Beads** 1/5 in Bead Binding Buffer (see Table in the corresponding section, add H2O to 50 ml)

Reverse Transcription

9

**Note**

When processing several plates for an experiment, try to balance the different conditions to be able to correct for potential batch effects. In the ideal case process all plates in one go. However, for the first steps until pooling of the cDNA you will need one Thermal Cycler per plate, so depending on the number of available Thermo Cyclers, batches can be bigger or smaller.

10 Prepare Reverse Transcription Mix

A	B	C
Reagent	Well	Plate
UltraPure Water	0.7 μ L	77 μ L
PEG 8000 (50% solution)	1.5 μ L	165 μ L
Maxima RT Buffer (5x)	2 μ L	220 μ L
dNTPs (25 mM)	0.4 μ L	44 μ L
dCTPs(100 mM)	0.2 μ L	22
TSO E5V7NEXT (100 μ M)	0.1 μ L	11 μ L
Maxima H Minus RT (200 U/ μ L)	0.1 μ L	11 μ L
Total	5 μ L	550 μ L

If you wish to add spike-in molecules like ERCCs or molecular spikes add them to the RT Mix and decrease the water accordingly

Note

Caution: Reverse Transcription Mix with PEG needs to be mixed carefully! RT Mix should be kept on ice until use but not longer than ⌚ 00:15:00 minutes

11 Gently thaw plates on ice for ⌚ 00:01:00 at most.

1m



12 Spin down in a pre-cooled centrifuge  1000 rcf, 4°C, 00:00:30 . 30s

13 Add  5 µL **Reverse Transcription Mix** to each well using a repeater pipette or a liquid handling robot like the Mantis.

14 Incubate for  01:30:00 at  42 °C in a Thermal Cycler with heated lid  105 °C .

cDNA Pooling & Purification

15 Mix each well (10 µL per well) with  10 µL of **Pooling Beads** for a 1:1 ratio. Pooling Beads can be added using a repeater pipette.

Note

The EDTA in the **Pooling Beads** will inactivate the RT and make pooling easier due to the color.

16 Pool all wells of one plate into a 2 mL tube or 15 mL tube when pooling up to 384 cells.

Note

We usually transfer all wells of a 96 well plate into the first column using a 20 µL 8-channel pipette and then transfer everything to a bigger tube using a 200 µL pipette.

17 Incubate for  00:05:00 at  Room temperature to allow binding of the first strand cDNA onto beads.

18 Place the tube on the magnet stand until clear, approximately  00:05:00 and 5m
discard supernatant.



- 19 Wash with  1 mL of **80% EtOH** while the tube is on the magnet. Discard the supernatant.

Note

Volume of EtOH should be adjusted depending on the number of cells/plates in a pool. More samples will require more EtOH to cover the beads completely and wash them efficiently.

- 20 Repeat wash step once more.

- 21 Air dry beads for  00:05:00

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

- 22 Resuspend the beads in  17 μ L of **UltraPure Water** and remove from Magnet.

- 23 Incubate for  00:05:00 at RT and place on magnet to transfer supernatant to a new well of a 96 well PCR plate.

5m

Exonuclease I Treatment

- 24 Add  2 μ L of **Exonuclease I Buffer (10x)** and  1 μ L of **Exonuclease I** per pool.

Note

The **Exonuclease I** step is important to remove remaining single stranded oligo dT primers. **When not removed completely oligo dT primers** will prime in the subsequent PCR and thus **lead to extremely high levels of cross contamination and inflation of UMI counts!**



25 Incubate as follows in a Thermal Cycler with heated Lid 🔥 105 °C :

Step	Temperature	Time
Incubation	37 C	20 min
Heat Inactivation	80 C	10 min
Storage	4 C	∞

Exonuclease I digest

26 Mix sample with 🧴 20 µL **Clean Up Beads (22% PEG)** for a ratio of 1:1

Note

This Clean up is necessary because the Exonuclease Reaction inhibits the Kapa HiFi polymerase. As this clean up is done on the pooled cDNA the loss of molecules is negligible.

27 Incubate for ⌚ 00:05:00 at 🔥 20 °C (Room Temp)

28 Place the plate on the magnet stand until clear (~ ⌚ 00:05:00) and discard supernatant.

5m

29 Wash with 🧴 100 µL of **80% EtOH** while the plate is on the magnet.

30 Discard the supernatant and keep plate on the magnet.

31 Repeat wash step once more. ➡ [go to step #29](#)

32 Air dry beads for ⌚ 00:05:00

**Note**

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

- 33 Remove the plate from the magnet and resuspend the beads in  20 μL of **UltraPure Water**.
- 34 Incubate for  00:05:00 at RT and place on magnet to transfer supernatant to a new well.

Full length cDNA Amplification

- 35 Prepare **Pre Amplification Mix**. Adjust to the number of pools.

Reagent	1x
KAPA HiFi 2x RM	25 μL
SINGV6 Primer (10 μM)	3 μL
UltraPure Water	2 μL
Total	30 μL

Pre Amplification PCR Master mix for one pool.

- 36 Add  30 μL **Pre Amplification Mix** to each pool.
- 37 Incubate the Pre Amplification PCR as follows:

	A	B	C	D
	Step	Temperature	Time	Cycles
	Initial Denaturation	98 C	3 min	1 cycle
	Denaturation	98 C	15 sec	15 cycles*
	Annealing	65 C	30 sec	
	Elongation	72 C	4 min	
	Final Elongation	72 C	10 min	1 cycle
	Storage	4 C	∞	

Note

Adjust the number of cycles based on input (cell number and expected amount of RNA per cell).
15 cycles is a good starting point for one full 96 well plate containing big cells like iPSCs.
For immune cells increase cycle number to 17-19 cycles.

cDNA Bead Purification

38 Mix sample with  40 µL **Clean Up Beads (22% PEG)** for a ratio of 1:0.8

39 Incubate for  00:05:00 at  20 °C (Room Temp)

40 Place the plate on the magnet rack until clear (~  00:05:00).

5m

41 Discard supernatant and keep plate on the magnet.

42 Wash with  100 µL of **80% EtOH** while the plate is on the magnet.

43 Discard supernatant and keep plate on the magnet.



44 Repeat wash step once more.  [go to step #42](#)

45 Air dry beads for  00:05:00

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

46 Remove from Magnet and resuspend the beads in  10 μL of **UltraPure Water**.

47 Incubate for  00:05:00 at RT and place on magnet to transfer supernatant to a new well.

Note

Safe Stopping Point! The Pre-Amplified cDNA can be stored for up to one week at  -20 °C . However usually we want to know if the experiment worked first and continue to do the concentration measurement and Bioanalyzer QC.

cDNA Quantification and Quality Check

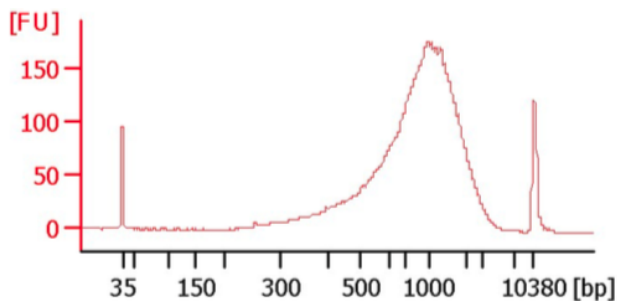
48 Quantify the cDNA using the **Quant-iT PicoGreen dsDNA assay kit** or equivalent following the manufacturer's protocol. Use 1 μL of clean cDNA for quantification.

49 Quality check the cDNA using the Agilent 2100 Bioanalyzer with **High Sensitivity DNA Analysis Kits**. Other instruments like the Tape Station or the Fragment Analyzer can be used as well but make sure to not waste more than 4 μL of your cDNA.

Note

Passing the cDNA quality check does not guarantee that the data will be of high quality, however, if the cDNA fails the quality check it will usually not yield good libraries and will therefore generate lower quality data.

Expected result



Exemplary cDNA trace with a **peak around 1000 to 1500 bp** and **few small fragments below 300 bp**.

50

Note

In previous versions of SCRB-seq/mcSCRB-seq we used the Nextera XT Kit for Library Preparation, but have switched to using NEB Next Ultra II recently. While the data quality is similar, the downscaled NEB Next Ultra II Kit as we use it is a fraction of the cost per library and allows for using more of the cDNA as input. However both methods can be used depending on availability of the kits in the lab.

Library Preparation

- 51 Use one fourth of your cDNA  2.5 μL but not more than  20 ng in total . If your cDNA concentration is higher than 8 ng/ μL , dilute accordingly.

**Note**

As the volumes in the scaled down NEBNext Ultra II kit are very low we recommend always preparing a 2x master mix for all subsequent steps.

Library Preparation - Fragmentation

- 52 Start by setting up the Thermo Cycler to be able to immediately proceed to the incubation after adding the Fragmentation Mix to the cDNA.

Note

Set heated lid to 75° C. Make sure the lid is at the correct temperature before you start the reaction.
Skip the first incubation step once you have added your samples.

	A	B	C
	Step	Temperature	Time
	Pre-Cool	4 °C	infinite
	Fragmentation	37 °C	5 min
	A Tailing and Phosphorylation	65 °C	30 min
	Storage	4 °C	infinite

- 53 Prepare **Fragmentation Mix**

	A	B
	Reagent	2x
	Ultra II FS Reaction Buffer	2.8 µL
	Ultra II FS Enzyme Mix	0.8 µL
	TE	3.4 µL
	Total	7 µL

Note

Ensure that the Ultra II FS Reaction Buffer is completely thawed. If a precipitate is seen in the buffer, pipette up and down several times to break it up, and quickly vortex to mix. Place on ice until use.

Vortex the Ultra II FS Enzyme Mix for 5-8 seconds prior to use for optimal performance.

54 Add  2.5 µL of cDNA (between 0.5 & 8 ng/µl) to a new well of a 96 well plate and add  3.5 µL of Fragmentation Mix.

55 Vortex the Fragmentation Mix for  00:00:05 and immediately proceed to next step.

5s

Safety information

Proceed to next step immediately to avoid over-fragmentation.

56 Place plate containing the samples into Thermal Cycler and start incubation by skipping the initial  4 °C hold.

Library Preparation - Adapter Ligation

57 Prepare Adapter Ligation Mix:

	A	B
Reagent	1 x	
Ultra II Ligation Master Mix	6 µL	
Ultra II Ligation Enhancer	0.2 µL	



	A	B
	prime Adapter (1.5 μ M)	0.5 μ L
	Total	6.7 μL

Adapter Ligation mix for one Library.

58 Add  6.7 μ L Adapter Ligation Mix to each replicate.

59 Incubate for  00:15:00 at  20 $^{\circ}$ C

15m

Note

Turn off heated lid or run cycler with open lid.

Library Preparation - Double size selection

60 Add  37.3 μ L Buffer EB to Samples for a total of  50 μ L

61 Mix Index PCR with  25 μ L SPRI select beads (ratio of 0.5x).

Note

We use SPRI Select Beads here instead of our home made 22% Clean Up beads for their guaranteed QCed size selection properties.

62 Incubate for  00:05:00 at room temperature.

5m



- 63 Place the plate on the magnet stand until clear and transfer  75 μL supernatant to a clean well.

Safety information

Be careful not to discard the supernatant! This is your library!

- 64 Mix supernatant with  10 μL SPRI select beads (ratio of 1:0.7)

- 65 Incubate for  00:05:00 at  Room temperature

5m

- 66 Place the plate on the magnet stand until clear.

- 67 Discard supernatant and keep plate on magnet.

- 68 Wash with  150 μL of 80% EtOH while the plate is on the magnet.

- 69 Discard the supernatant and keep plate on magnet.

- 70 Repeat wash step once more.  [go to step #68](#)

- 71 Air dry beads for  00:05:00

5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.



72 Take plate off the magnet and resuspend samples in  10.5 μL 0.1X TE (dilute 1X TE Buffer 1:10 in water).

73 Elute for  00:05:00 minutes.

5m

74 Place plate on magnet and transfer samples to clean wells.

Library Preparation - Library PCR and Indexing

75

Note

For Illumina sequencers with patterned flow cells (e.g. Nova-seq or Next-seq Series) it is recommended to use unique dual indexing, meaning both the i5 and i7 indices are used for only one library. Library replicates may have the same indices.

76 Add  1 μL of Nextera i7 Index Primer ( 5 micromolar (μM)) to each well.

77 Add  1 μL of TruSeq i5 Index Primer ( 5 micromolar (μM)) to each well.

78 Add  12.5 μL of 2x Q5 Master Mix (NEBNext Ultra II) to each well.

Note

Although scaled down, there will not be sufficient Q5 Master Mix (M0544L) in the kit. This item will have to be ordered separately.

79 Incubate Library PCR reaction as follows with the heated Lid set to  105 $^{\circ}\text{C}$:



	A	B	C	D
	Step	Temperature	Time	Cycles
	Initial Denaturation	98 °C	30 sec	
	Denaturation	98 °C	10 sec	10*
	Annealing/Elongation	65 °C	1 min	10*
	Final Elongation	65 °C	5 min	
	Storage	8 °C	∞	

Library Amplification PCR

- 80 Adjust the number of cycles based on total cDNA input.
As a general guide we recommend:

A	B
Input (ng)	Cycles
20	10
10	11
5	12
2.5	13

Library Preparation - Final Double Size Selection and Clean Up

- 81 Add  25 µL Buffer EB to Index PCR.
- 82 Mix Index PCR with  25 µL SPRI select beads (ratio of 1:0.5)
- 83 Incubate for  00:05:00 at  Room temperature

5m



- 84 Place the plate on the magnet stand until clear and transfer  75 μL supernatant to a clean well.

Safety information

Be careful not to discard! This is your library.

- 85 Mix supernatant with  10 μL SPRI select beads (ratio of 1:0.7)

- 86 Incubate for  00:05:00 at  Room temperature .

5m

- 87 Place the plate on the magnet stand until clear.

- 88 Discard supernatant and keep plate on the magnet.

- 89 Wash with  150 μL of 80% EtOH while the plate is on the magnet.

- 90 Discard supernatant and keep plate on the magnet.

- 91 Repeat wash step once more.  [go to step #89](#)

- 92 Air dry beads for  00:05:00 .

5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.



- 93 Elute in 15 μ l UltraPure Water. 107 Incubate for 00:05:00 and then place on magnet until clear. Transfer eluted library to new well. Stopping point. The libraries can be safely stored at -20°C until they will be QCed and sequenced. Library QC 45m
- 94 Take plate off the magnet and resuspend samples in  15 μ L **UltraPure Water**.
- 95 Elute for  00:05:00 minutes.
- 96 Transfer  15 μ L clean sequencing Library to a 0.5 mL tube for storage.

Note

Safe Stopping Point! The final library can be stored at  -20°C until sequencing ideally not longer than one month.

Library Preparation - Quantification and QC

- 97 Quantify the cDNA using the **Quant-iT PicoGreen dsDNA assay kit** or equivalent following the manufacturer's protocol. Use  1 μ L of the final Library for quantification.

Note

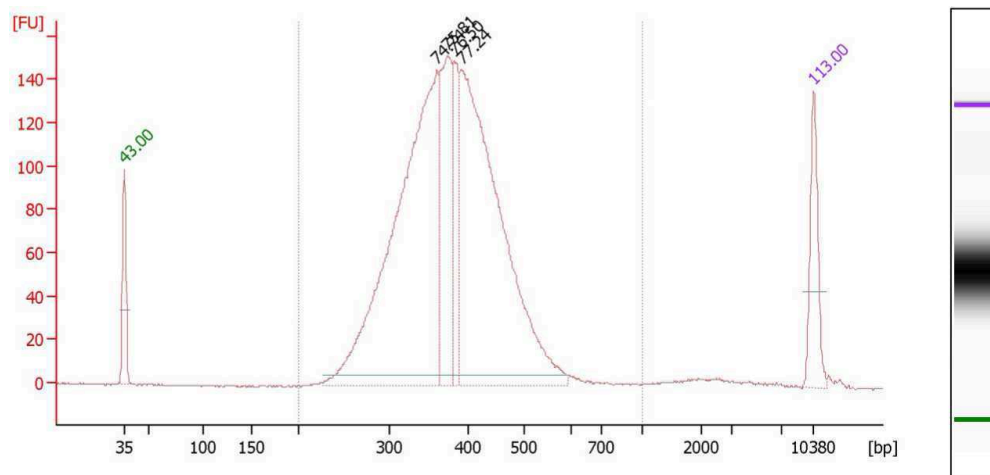
Expected Library concentrations are between 1 and 10 ng/ μ l. Very low yield can be an indicator of low quality cDNA. For Yields between 0.1 and 1 ng/ μ l consider Amplifying the Library for 5 to 10 more Cycles using the Q5 Master Mix with universal Illumina Fwd and Rev. primers. (see Materials)

- 98 Quality control and Quantify the Library using the Agilent 2100 Bioanalyzer with **High Sensitivity DNA Analysis Kits**. Other instruments like the Tape Station or the Fragment Analyzer can be used as well.

Note

Use Library concentrations between 1 and 2 ng/μL for optimal quantification results. If Library concentration is above 2 ng/μl dilute accordingly.

Expected result



Optimal Bioanalyzer trace of an NEB Next Library. Narrow distribution with a peak between 300 and 400 bp. If the Fluorescence intensity maximum is markedly higher than the Marker peaks (35bp and ~10000 bp) the quantification of the Bioanalyzer will be inaccurate. The trace in the example is still in a quantifiable range.

- 99 Quantify the Library Molarity using the Bioanalyzer2000 (or else) software. Make sure to set the Region from 200 bp to 1000 bp. The Bioanalyzer will calculate the Molarity based on fragment size and Fluorescence intensity. We usually aim for a Library concentration of 10nM (10000pM) to submit for sequencing. Lower or higher molarities might be required depending on the sequencing provider.

Sequencing

- 100 Samples should be submitted according to your Sequencing Facility specifications. prime-seq is compatible with Illumina Sequencing.

At least 8 cycles are required for the Index Reads (i7+i5) and 28 cycles for the Read 1 (BC+UMI). Read 2 (DNA) should be adjusted based on the quality of the genome being mapped to, but for human and mouse 50 cycles are sufficient.

Some potential sequencing options:

	A	B	C	D	E	F
	Sequencer	Read1	Read2	Index Read (i7)	Index Read (i5)	Kit
	NovaSeq 6000	28	94	8	8	SP v1.5 100 cycle
	NovaSeq 6000	150	150	10	10	S4 v1.5 300 cycle
	NextSeq 500/550	28	63	8	8	NextSeq 500/550 HiOut v3 75 cycle
	NextSeq 1000/2000	28	88	8	8	NextSeq 1000/2000 P2 100 cycle
	NextSeq 2000	28	46	8	0	NextSeq 2000 P3 50 cycle

NextSeq 2000 P3 50 Cycle is only possible when not pooling with other libraries as no index read is included.

- 101 Sequencing Depth should be adjusted to the scientific question, for example for broad cell type classification few read per cell between 10 k and 25 k are usually sufficient. For in depth transcriptome analysis between 100k - 500k reads per cell are adequate. Please note that these are just general remarks and library complexity may differ considerably depending on cell type, state and quality of the input.

	A	B	C
	Number of cells	Mio. reads shallow seq. (25k per Cell)	Mio. reads deep seq. (500k per Cell)
	96	2.4	48



	A	B	C
	384	9.6	192
	1536	38.4	768

Exemplary sequencing depth calculations.

Appendix: Prepare Clean Up Beads (SPRI 22% PEG)

102 Prepare **PEG Solution (22%)** by adding all ingredients to a 50 mL falcon tube

	Reagent	Amount
	PEG 8000	11 g
	NaCl (5M)	10 mL
	Tris-HCl (1M, pH 8.0)	500 µL
	EDTA (0.5M)	100 µL
	IGEPAL (10% solution)	50 µL
	Sodium Azide (10% solution)	250 µL
	UltraPure Water	up to 49 mL
	Total	49 mL

Bead Binding Buffer

Note

Do not add the total amount of water until after PEG is completely solubilized

103 Incubate at  40 °C and vortex regularly until PEG is completely dissolved



- 104 Resuspend **Sera-Mag Speed Beads** carefully and pipette  1000 μL of bead suspension into a 1.5 mL tube
- 105 Place on magnet stand and remove storage buffer
- 106 Add  1000 μL of **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads
- 107 Place on magnet stand and remove supernatant
- 108 Repeat wash step one more time
- 109 Add  900 μL **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads
- 110 Add the washed **Sera-Mag Speed Beads** to the **PEG Solution (22%)** and mix well

Note

The final **Cleanup Beads** can be aliquoted and stored at  4 °C for up to six months

Appendix: Prepare Pooling Beads (SPRI 22% PEG 5 times less Beads)

- 111 Prepare **PEG Solution (22%)** by adding all ingredients to a 50 mL falcon tube

	Reagent	Amount
	PEG 8000	11 g
	NaCl (5M)	10 mL
	Tris-HCl (1M, pH 8.0)	500 μL
	EDTA (0.5M)	100 μL



	IGEPAL (10% solution)	50 µL
	Sodium Azide (10% solution)	250 µL
	UltraPure Water	up to 49 mL
	Total	49 mL

Note

Do not add the total amount of water until after PEG is completely solubilized

- 112 Incubate at  40 °C and vortex regularly until PEG is completely dissolved
- 113 Resuspend **Sera-Mag Speed Beads** carefully and pipette  200 µL of bead suspension into a 1.5 mL tube
- 114 Place on magnet stand and remove storage buffer
- 115 Add  1000 µL of **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads
- 116 Place on magnet stand and remove supernatant
- 117 Repeat wash step one more time
- 118 Add  900 µL **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads



Appendix: Prepare Pooling Beads (SPRI 22% PEG 5 times less Beads)

- 119 Add the washed **Sera-Mag Speed Beads** to the **PEG Solution (22%)** and mix well

Note

The final **Cleanup Beads** can be aliquoted and stored at  4 °C for up to six months